Water

Sources and Impurities in Water

- **Sources**
- **Impurities**

i) Physical Impurities

ii) Chemical Impurities

iii) Biological Impurities

Acidity in water

$H_2O + CO_2 \rightarrow H_2CO_3$

Hardness of Water

Hardness of water is that characteristic which prevents the lathering of Soaps.

Or’

It is the soap consuming capacity of water sample.

Industrial problems of water

- Boiler corrosion
- Caustic embrittlement
- Formation of scale and sludge
- Priming and foaming

TEXTILE INDUSTRY

- Hard water causes much of soap used in washing yarn, fabric etc. to go as waste because hard water cannot produce good quality of lather.
- Precipitates of **calcium** and **magnesium** soaps adhere to fabrics.
- These fabrics when dyed doesn’t produce exact shades of color.
- If the water contains **Iron, manganese, color or turbidity**, it leaves stains on fabrics.
- Organic matter imparts foul smell.



SUGAR INDUSTRY

- Water containing **sulphates**, **nitrates**, **alkali carbonated**, etc., if used in sugar refining causes difficulties in the crystallization of sugar.
- Moreover, the sugar so produced may be deliquescent.
- The color and taste of the sugar also changes.



DYEING INDUSTRY

- Hard water contains dissolved **calcium**, **magnesium** and **iron** salts. These salts present in hard water may react with costly dyes, forming undesirable precipitates, which yield impure shades.
- Uniform dyeing is not possible because

1. Hard water decreases the solubility of dye or color. (PH increases)

2. Basic dyes also precipitate out in hard water. (eg. Ca^{2+} ions react with bicarbonate.

(HCO_3^-) ions to form insoluble calcium carbonate ($CaCO_3$))

- They also give spots on the fabrics being dyed.

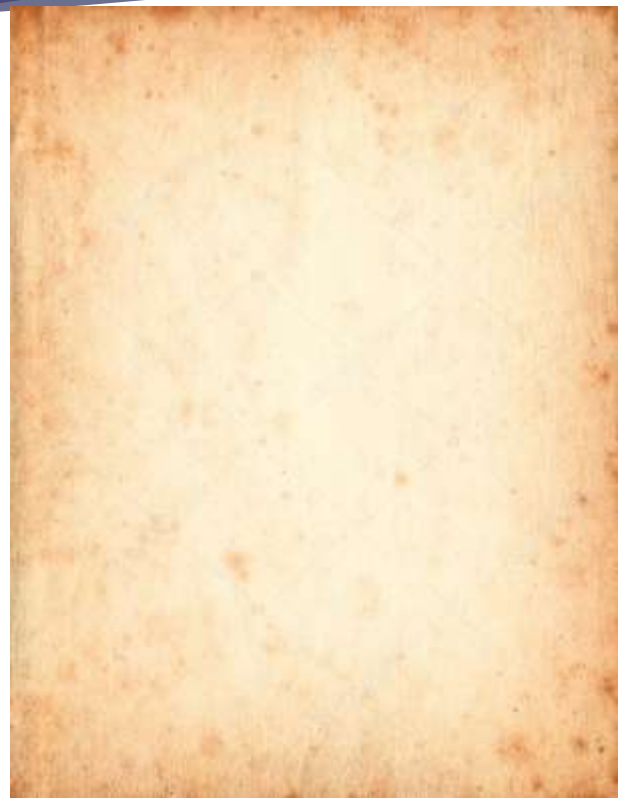


PAPER INDUSTRY

- **Calcium** and **magnesium** salts present in hard water tend to react with chemicals and other materials employed to provide a smooth and glossy (i.e. shining) finish to paper.
- It also produces in the cracks in paper.
- Alkalinity (consumes alum and increases the cost of production)

REACTION: $Al_2(SO_4)_3.14H_2O + 6HCO_3^- \rightarrow 2Al(OH)_3 + 6CO_2 + 14H_2O + 3SO_4^{2-}$

- Moreover, **iron** and **manganese** salts may even affect the brightness and color of paper being produced.
- **Calcium** and **magnesium** salts increase the ash content of paper.



LAUNDRY INDUSTRY

- Laundries require soft water, free from color, **manganese** and **iron**
- Due to hard water, consumption of soaps increases.
- **Iron** and **manganese** salts impart a grey or yellow shade to the fabric.
- Weakening of fibres causing tears and thus shortening the life of the fabric.
- Fabrics lose their soft and supple feel to become stiff and harsh.
- Extra detergent, extra rinse and wash cycles to clean the laundry.



PHARMACEUTICAL INDUSTRY

- Pharmaceutical industries require ultra pure water which should be colorless, odourless, tasteless and free from pathogenic organisms.
- Hard water, if used for preparing pharmaceutical products like drugs, injection, ointments, etc. may produce undesirable products in them.
- According to the pharmaceutical guidelines stated by

WORLD HEALTH ORGANISTION

If the pharmaceutical manufacturer's water supply is “**hard**” it needs to be “**softened**” which increases the cost of production.

PHARMACEUTICAL INDUSTRY

Application Scheme



BEVERAGE INDUSTRY

- Water used in beverage industries is very sensitive.
- Water should be free from any contaminants and micro-organisms.
- Water required should not be alkaline, as it destroys or modifies the taste as it tends to neutralize the fruit acids.
- If hard water is used in this industry, it can change soft drink color, test and odor. Moreover a deposit can be developed here.
- So water must be treated before use for decreasing the effects of hard water in beverages industries. Otherwise it would lessen the product quality.

RUST FORMATION IN PIPES

- The carbonate hardness salts cause deposition in tubes and pipes:

REACTION: $Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$

- The calcium carbonate (**$CaCO_3$**) thus formed is responsible for deposition in pipes.
- Similarly the **CO_2** which is released reacts with water to form carbonic acid which is corrosive and causes severe corrosion in pipes.
- This decreases the strength of the material making it unsafe to bear high temperature and pressure.



SLUDGE AND SCALE FORMATION

- In **boilers**, conc. of salts increases due to continuous evaporation of water.
- When the conc. reaches saturation point it is thrown out of the water in form of ppt, which gets settled on the inner walls of the boiler.
- If the ppt takes place in form of loose and soft ppt it is called as **sludge**. It is formed at comparatively colder regions of boiler or at bends. Eg. **$Mg(OH)_2$**
- If the ppt sticks firmly to the inner walls of the boiler it called as **scales**. Sometimes it is even difficult to remove with hammer or chisel. Eg. **$MgSiO_3$**
- **DISADVANTAGES OF SLUDGE AND SCALE FORMATION**

1. Wastage of fuel
2. Lowering of boiler safety
3. Decrease in efficiency
4. Danger of explosion

SLUDGE AND SCALE FORMATION

SCALE

SLUDGE





CAUSTIC EMBRITTLEMENT

- In a boiler, some qty. of Na_2CO_3 added to soften the water. But since high pressure and high temperature in the boiler converts Na_2CO_3 to NaOH.

  REACTION: Na2CO3 + H2O → 2NaOH + CO2

- Because of NaOH, boiler water becomes alkaline. This very dilute alkaline water flows through minute hair cracks.
- When water evaporates, this dilute alkali becomes concentrated.
- This concentrated alkali reacts with **iron** to form **sodium ferroate.**

  REACTION: $2NaOH + Fe \rightarrow Na_2FeO_2 + H_2$

- This sodium ferroate gets deposited in crevice and cracks.
- This reaction is reversible i.e. where at high temperature sodium ferroate decomposes to form NaOH and Fe_3O_4 . So reaction continues making use of iron metal of boiler.
- **REACTION: $Na_2FeO_2 + H_2O + O_2 \rightarrow 12NaOH + 2\ Fe_3O_4$**



1.Engineering Chemistry Textbook by Chopra and Parmar
2.Engineering Chemistry by Pearson
3.http://textilelearner.blogspot.com/
4.http://www.environmentalpollution.in/
5.https://www.chegg.com/homework-help/
6.http://web.deu.edu.tr/atiksu/ana52/
7.http://apps.who.int/medicinedocs/

THANK YOU!

Soap and Soap Consuming capacity

Soap is sodium or potassium salt of higher fatty acids like oleic, palmitic or steric acid.

Soap Consuming capacity of water

$C_{17}H_{35}COONa + CaCl_2 \rightarrow (C_{17}H_{35}COO)_2Ca + 2NaCl$

Sodium stearate (Soap) Hardness Calcium stearate (white Scum)

❑ Hard Water & Soft water

- The water which does not produce lather with soap solution readily is called as hard water.

- Water which lathers easily on shaking with soap solution is called soft water.

Types of hardness

- **Temporary or Carbonate or Alkaline Hardness**

The hardness caused by the presence of dissolved bicarbonates and carbonates of calcium, magnesium and other heavy metals like iron is called as temporary hardness. Eg. $CaCO_3$, $Ca(HCO_3)_2$, $MgCO_3$, $Mg(HCO_3)_2$, $FeCO_3$, $NaHCO_3$

-non-hardness causing substance, SiO_2, NaCl, KCO_3, $KHCO_3$

- **Permanent or non-carbonate or non-alkaline hardness**

The hardness caused by the presence of chlorides and sulphates of calcium, magnesium, iron and other heavy metals. eg. $CaCl_2$, $MgCl_2$, $FeCl_3$, $CaBr_2$, $CaSO_4$, etc

KBr-non-hardness causing substance.

- Temporary Hardness can be destroyed by boiling.
- boil

$Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$

Insoluble

$Mg(HCO_3)_2 \rightarrow Mg(OH)_2 + CO_2$

Insoluble

Permanent Hardness cannot be destroyed by boiling.

1. $KHCO_3$
2. $Mg(NO_3)_2$= 10ppm $CaCO_3$ eq. =3.5ppm
3. $FeSO_4$= 12ppm $CaCO_3$= 2.4ppm
4. $NaAlO_2$
5. $CaCl_2$= 7ppm $CaCO_3$ = 1.1ppm
6. $Ca(NO_3)_2$ =60ppm $CaCO_3$ eq. = 50ppm

Hardness of water ($CaCO_3$ eq) = 3.5+2.4+1.1+50 =57ppm

$CaCO_3$ is chosen as basis to express hardness of water.

Measurement of Hardness

- The extent of hardness is measured in terms of ions contributing to the hardness.
- It is expressed in terms of amount of $CaCO_3$

- Why to express hardness in terms of CaCO3?
- Ans: 1. the mol wt of CaCO3 is 100 and its equivalent wt is 50, which allows the multiplication and division in easy way when required.
  2. CaCO3 is the most insoluble salt which can be precipitated in water treatment.

The Equivalent of $CaCO_3$

= Mass of hardness producing substance x multiplication factor(M.F.)

Where Multiplication factor

$$= \frac{\text{Chemical eq. of } CaCO_3}{\text{Chemical Eq. of hardness producing substance}}$$

= 50/ 74 {for $Mg(NO_3)_2$}

Multiplication factors

Dissolved salt	Molar Mass	Chem. Eq.	M.F
1. Ca(HCO3)2	162	81	50/81 or 100/162
2. Mg(HCO3)2	146	73	50/73 or 100/146
3. CaSO4	136	68	50/68 or 100/136
4. FeCl2	127	63.5	50/63.5 or 100/127
5. CaCl2	111	55.5	50/55.5 or 100/111
6. MgSO4	120	60	50/60 or 100/120
7. MgCl2	95	47.5	50/47.5 or 100/95
8. CaCO3	100	50	1
9. MgCO3	84	42	50/42 or 100/84
10. CO2	44	22	50/22 or 100/44
11. Ca(NO3)2	164	82	50/82 or 100/164
12. Mg(NO3)2	148	74	50/74 or 100/148
13. HCO_3^-	61	61	50/61 or 100/122
14. OH-	17	17	50/17 or 100/34

Dissolved salt	**Molar Mass**	**Chem. Eq.**	**M.F**
15. CO_3^{2-}	60	30	50/30 or 100/60
16. NaAlO2	82	82	50/82 or 100/164
17. Al2(SO4)3	342	57	50/57 or 100/114
18. FeSO4.7H2O	278	139	50/139 or 100/278
19. H+	1	1	50/1 or 100/2
20. HCl	36.5	36.5	50/36.5 or 100/73

Units of hardness

1. Parts per millions(ppm): It is the parts of $CaCO_3$ equivalent of hardness per10^6 parts of water.
2. Milligram per Litre(mg/L): it is the number of milligram of CaCO3 equivalent of hardness present per litre of water.

Mg/L = 1gx10^-3/1000mL= 1g/1000x1000 = 1:10^6

3. Clarke's Degree (°Cl): It is the number of grains (1/70000 lbs) of CaCO3 equivalent of hardness per gallon (10 lbs) of water. OR It is parts of CaCO3 equivalent per 70000 parts of water.
4. Degree French (°Fr): It is the parts of CaCO3 equivalent per 10^5 parts of water.
5. Milliequivalent per Litre(meq/L): It is the number of milliequivalent of hardness present per litre of water.

Relationship between various units of hardness

1ppm = 1mg/L = 0.1°Fr = 0.07°Cl = 0.02 meq/L

Numerical 1: Convert the total hardness of water samples in ppm and meq/L from the following.

i) 20.23°Cl

ii) 31.3°Fr

Numerical 2: A sample of water has hardness 208ppm of $CaCO_3$ equivalents. Find the hardness in terms of

i) mg/L ii) °Fr iii) °Cl iv) meq/L

Numerical 3: How many grams of $FeSO_4$ dissolved per litre gives 210.5ppm of hardness? (Fe = 56, S =32, O =16, Ca =40, C = 12)

Soln: Mol. Wt of $CaCO_3$ = 100g and mol. Wt of $FeSO_4$ = 152g

100g of $CaCO_3$ = 152g of $FeSO_4$

100ppm of $CaCO_3$ = 152ppm of $FeSO_4$

Hence, 210.5ppm of hardness = 152 x 210.5/100 ppm of $FeSO_4$

= 319.96 mg/L

=0.320g/L of $FeSO_4$

Numerical 4: Calculate Temporary hardness and permanent hardness of water sample from the following data: $Mg(HCO_3)_2$ = 16.8mg/L, $MgCl_2$ = 19mg/L, $MgSO_4$ = 24mg/L, $Mg(NO_3)_2$ = 29.6mg/L, $CaCO_3$ = 4mg/L, $MgCO_3$ = 10mg/L.

Solution:

Constituent (mg/L)	Multiplication Factor	$CaCO_3$ Eq.	Hardness
1. $Mg(HCO_3)_2$ = 16.8	50/73	11.5	T
2. $MgCl_2$ = 19	50/47.5	20	P
3. $MgSO_4$ = 24	50/60	20	P
4. $Mg(NO_3)_2$= 29.6	50/74	20	P
5. $CaCO_3$ = 4	1	4	T
6. $MgCO_3$ = 10	50/42	11.9	T

Temporary Hardness = hardness due to $Mg(HCO_3)_2$ +$CaCO_3$ + $MgCO_3$

= 11.5 + 4 + 11.9 = 27.4 ppm or 27.4 mg/L

Permanent Hardness = hardness due to $MgCl_2$ + $MgSO_4$ + $Mg(NO_3)_2$

= 20 + 20+ 20 = 60mg/L = 60ppm

Numerical 5: A water sample analysis has been found to contain following impurities in ppm: $Mg(HCO_3)_2$ =14.6, $Mg(NO_3)_2$ = 29.6, $Ca(HCO_3)_2$= 8.1, $MgCl_2$ = 19, $MgSO_4$ = 24. calculate the temporary & permanent hardness of water.

Numerical 6: How many grams of $CaCl_2$ dissolved per litre gives 150ppm hardness?

Numerical 7: Classify the following impurities into temporary permanent and non-hardness causing impurities.

$Ca(HCO_3)_2$, $MgSO_4$, $CaCl_2$, CO_2, HCl, $Mg(HCO_3)_2$

Determination of hardness of water by EDTA method

- Hardness of water can be readily determined by titration with the chelating agent EDTA.
- EDTA is ethylene diamine tetra acetic acid. It is a weak acid that can lose four H+ ions on complete neutralization. Its formula is

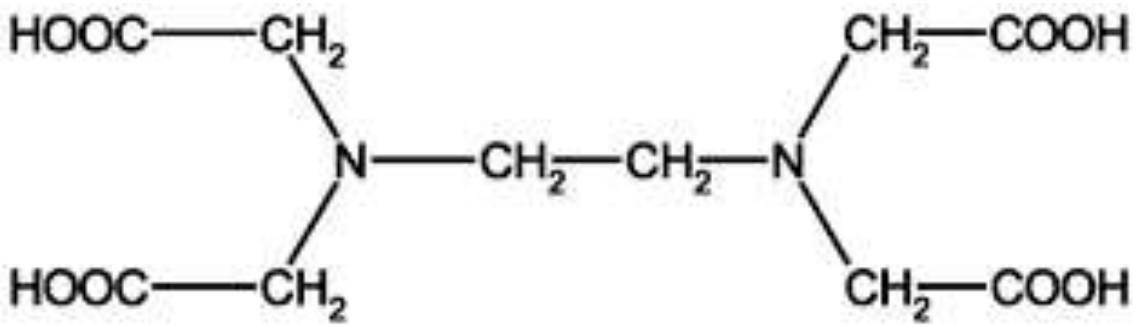
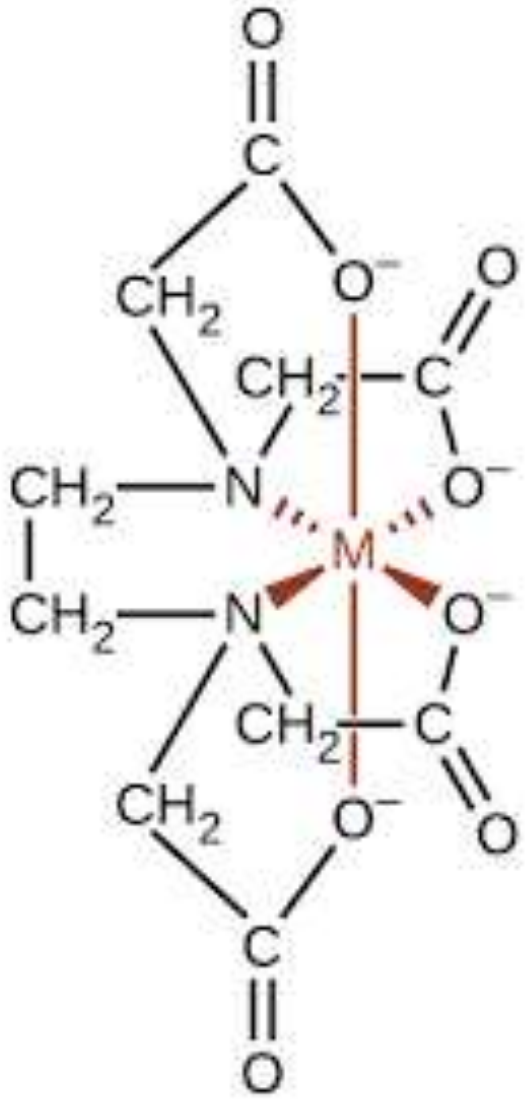


Figure 1. Molecular structure of EDTA

- M-EDTA Complex structure: Stable at pH=10 Octahedral complex



Theory or Principle:

- The process involves the titration of metal ion solution with standardized EDTA soln. Indicator used is Eriochrome Black-T (EBT).
- EBT is alcoholic solution of blue eye. It forms an unstable wine-red complex with Ca2+ & Mg2+ at pH =10. A buffer solution of NH4OH & NH4Cl is used to maintain the pH = 10.
- **Initial reaction:** When EBT indicator is added to hard water buffered to pH =10, a wine red unstable complex is formed.

pH = 10

M^{2+} + EBT → [M-EBT] complex (Unstable)

(Wine red)

Titration Reaction: During the course of titration with EDTA soln, EDTA combines with M2+ ions to form stable complex M-EDTA & releases free EBT indicator, which immediately combines with remaining M2+ ions present in solution. Thus wine red colour retained.

[M-EBT] complex + EDTA → [M-EDTA] complex + EBT

pH = 10

M^{2+} + EBT → [M-EBT] complex

End point Reaction: At end point all of the cations formed M-EDTA complex, where last drop of EDTA added displaces the EBT indicator from M_EBT complex and wine red colour changes to blue due to presence of only free EBT. Thus the colour change is wine red to blue at the end point of titration.

pH = 10

[M-EBT] complex + EDTA → [M-EDTA] complex + EBT

Colour change : Wine red to Blue

Practical Colour change: Wine red to purple to blue

Experimental Procedure:

A) Preparations of solutions

i) Preparation of Standard Hard water (SHW):

Dissolve 1g of $CaCO_3$ in minimum amount of dil. HCl till all solid dissolves and evaporated to dryness. Dissolve this residue in 1L distilled water. Each mL of this solution contains 1mg of $CaCO_3$ equivalent hardness.

1g of $CaCO_3$ in 1L of water= 1000mg of $CaCO_3$ in 1L = 1000ppm soln

1mL of SHW ($CaCO_3$) soln = 1mg of $CaCO_3$

100mL of SHW = _100__mg of $CaCO_3$

1ppm = 1mg/L

ii) Preparation of EDTA solution: Dissolve 3.72g of pure EDTA + 0.1 mg $MgCl_2$ in 1L of distilled water.

iii) Preparation of indicator: dissolve 0.5g of EBT indicator in 100ml of alcohol.

iv) Preparation of Buffer soln: Add 67.5g of NH_4Cl to 570mL of ammonia soln and then dilute with distilled water to 1L.

B) **Procedure: it has three steps:**

Step 1: Standardization of Hard water: Pipette out 50mL of SHW in a conical flask. Add 10mL of buffer soln and 4 to 5 drops of EBT indicator. Titrate with EDTA soln till wine red colour changes to blue. Let the volume used be V1 mL.

50mL of SHW = V1 mL of EDTA

1ml of EDTA = ___mg of $CaCO_3$

Step 2: Titration of unknown hard (bore well) water: Titrate 50mL of sample water just same as given instep 1. Let the volume used be V2 mL.

50mL of sample HW = V2 mL of EDTA

Total Hardness

Step 3: Titration of permanent hardness: Take 50mL of water sample in a beaker and boil it till volume reduces to half. (where all bicarbonates are decomposed to insoluble $CaCO_3$ and $Mg(OH)_2$.) Filter the solution and titrate the filtrate as given in step 1. Let the volume used be V3 mL.

50ml of boiled sample HW = V3 mL of EDTA

Permanent Hardness

Calculations:

50ml of SHW = V1 mL of EDTA soln

Since 1 mL SHW contains 1mg of $CaCO_3$,

50mg of $CaCO_3$ = V1 mL of ETDA soln.

Hence, **1mL of EDTA = 50/V1 mg of $CaCO_3$ eq.**

A) Total Hardness

50mL of sample hard water = V2 mL of EDTA soln

= V2 x 50/V1 mg of $CaCO_3$ eq.

Hence 1000mL of sample hard water = 1000 V2x 50/V1 x 1/50 mg of $CaCO_3$ eq.

there fore, **Total Hardness = V2/V1 x 1000 ppm**

B) Permanent Hardness:

Now, 50ml of boiled sample water = V3 mL of EDTA

= V3 x 50/V1 mg of $CaCO_3$

Hence 1000mL of Sample hard water = V3/V1 x 1000 mg of $CaCO_3$

Therefore, Permanent hardness = V3/V1 x 1000 ppm

C) Temporary hardness = Total hardness – Permanent hardness

Advantages of EDTA method

- This method is preferable to the other methods, because of

i) Greater accuracy

ii) Convenience

iii) More rapid procedure

Numerical 1: 50mL of standard hard water requires 38mL of EDTA solution. 100 mL of water sample consumes 21mL of EDTA soln. 100mL of boiled water sample consumes 10mL of EDTA soln. Find all types of hardness.

Solution:

50 mL of SHW = _50_mg of $CaCO_3$

50mL of SHW = 38 ml of EDTA soln.

There fore, 38mL of EDTA = 50mg of $CaCO_3$

1mL of EDTA = _50/38__mg of $CaCO_3$

1mL of EDTA = 1.316mg of $CaCO_3$

Total Hardness:

100mL of sample HW =21 mL of EDTA solution

= 21 x 1.316 mg of $CaCO_3$

= 27.636 mg of $CaCO_3$

1000mL of sample HW = 27.636 x 1000/100

= 276.36mg of $CaCO_3$/L = 276.36ppm

There fore **Total hardness = 276.36ppm**

Permanent hardness:

100mL of Boiled sample HW = 10mL of EDTA

= 13.16 mg of $CaCO_3$

1000mL of boiled sample HW = 13.16 x 1000/100 = 131.6mg of $CaCO_3$ / L

Permanent hardness = 131.6ppm

Temporary Hardness = Total hardness – permanent hardness

Numerical 2: 50 mL of Standard hard water required 21mL of EDTA solution. 50mL of sample Water requires 21mL of EDTA solution. 50mL of boiled sample water requires 7mL of EDTA soln. calculate all the types of hardness.

Soln:

50mL of SHW = 50mg of $CaCO_3$

50mL of SHW = 21mL of EDTA

There fore 21 mL of EDTA soln = 50mg of $CaCO_3$

Hence **1mL of EDTA soln = 50/21 = 2.381mg of $CaCO_3$ eq.**

Total Hardness

50mL of sample HW = 21mL of EDTA soln

= 21 x 2.381 = 50.001 mg of $CaCO_3$ eq.

1000mL of sample HW = 50.001 x 1000/50 = 50.001 x 20

= 1000.02mg $CaCO_3$/L

Hence Total hardness = 1000.02ppm

Permanent Hardness

50mL of boiled sample HW = 7mL of EDTA soln

= 7 x 2.381 = 16.667mg of $CaCO_3$/L

1000mL of boiled sample HW = 16.667 x 1000/50 = 16.667 x 20 =333.34mg of $CaCO_3$

Hence permanent hardness = 333.34ppm

Temporary Hardness = total hardness – permanent hardness = 1000.02 -333.34 = 666.68 ppm

Numerical 3: 0.38g of $CaCO_3$ was dissolved in HCl and the solution made up to 1Litre with distilled water. 100 mL of the above soln required 25mL of EDTA soln. 100 mL of hard water sample required 37mL of EDTA soln. 100mL of boiled hard water sample required 10mL of EDTA soln. Calculate each type of hardness.

Soln

~~1ml of SHW = 1mg of CACO3~~

SHW = 0.38g of $CaCO_3$/ 1000mL = 380mg of $CaCO_3$ / 1000mL

1mL of SHW = 0.38mg of $CaCO_3$……using 1st statement of numerical

100mL of SHW = 25mL of EDTA

100mL of SHW = 38 mg of $CaCO_3$

There fore 25mL of EDTA = 38mg of $CaCO_3$

Hence **1mL of EDTA = 38/25 =1.52mg of $CaCO_3$ ……using 2nd statement of numerical**

Total Hardness

100mL of sample HW = 37mL of EDTA

= 37 x 1.52 = 56.24 mg of $CaCO_3$

1000ml of sample HW = 56.24 x 1000/100 = 562.4 mg of $CaCO_3$ /L

Hence Total hardness = 562.4ppm

Permanent hardness

100mL of Boiled sample HW = 10 ml of EDTA

= 10 x 1.52 = 15.2 mg of $CaCO_3$

1000mL of Boiled sample HW = 15.2 x 1000/100 = 152 mg of $CaCO_3$

Hence permanent hardness = 152ppm

Temporary Hardness = total Hardness – permanent hardness = 562.4-152 = 410.4ppm

Numerical 4: 1.2g of $CaCO_3$ was dissolved in HCl and the solution made up to 1Litre with distilled water. 50 mL of the above soln required 30 of EDTA soln. 100 mL of hard water sample required 27mL of EDTA soln. 100mL of boiled hard water sample required 11mL of EDTA soln. Calculate each type of hardness.

Soln:

Numerical 5: 0.55g of $CaCO_3$ was dissolved in HCl and the solution made up to 1Litre with distilled water. 20 mL of the above soln required 10 of EDTA soln. 50 mL of hard water sample required 17mL of EDTA soln. 10mL of boiled hard water sample required 1.1mL of EDTA soln. Calculate each type of hardness.

Ans:

Numerical 6: 50mL sample of water require 10.2mL of M/20 disodium EDTA for titration. After boiling and filtration the same volume required 6.5mL of EDTA. Calculate all the types of hardness in it.

Numerical 6: 50mL sample of water require 10.2mL of M/20 disodium EDTA for titration. After boiling and filtration the same volume required 6.5mL of EDTA. Calculate all the types of hardness in it.

Soln:

1mL of 0.05M EDTA = ___mg of $CaCO_3$

M/20 EDTA = 1/20M EDTA = 0.05M EDTA

Std relationships says that,

1000mL of 1M EDTA = 100g of $CaCO_3$……due to Avogadro's numbers of molecules

1mL of 1M EDTA = 100mg of $CaCO_3$

1mL of 0.05M EDTA = 100 x 0.05 = 5mg of $CaCO_3$

Total Hardness:

50 mL of sample HW = 10.2mL of 0.05M EDTA soln

= 10.2 x 5 =51mg of $CaCO_3$ eq.

1000mL of sample HW = 51 x 1000/50 = 51 x 20 = 1020mg of $CaCO_3$

Hence Total Hardness = 1020ppm

Permanent Hardness:

50mL of Boiled sample HW = 6.5mL of 0.05M EDTA

= 6.5 x 5 = 32.5mg of $CaCO_3$ eq.

1000mL of boiled Sample HW = 32.5 x 20 = 650mg of $CaCO_3$ eq

Hence Permanent Hardness = 650ppm

So, temporary hardness = 1020-650 = 370ppm

Numerical 7: 50mL sample of water require 15.3mL of M/15 disodium EDTA for titration. After boiling and filtration the same volume required 9.5mL of EDTA. Calculate all the types of hardness in it.

Solution

How to soften the hard water?

- Lime – soda method (treatment with $Ca(OH)_2$ and Na_2CO_3)
- Zeolite method ($Na_2O.\ Al_2O_3.xSiO_2.yH_2O$)
- Ion exchange method (CO-Polymers of styrene and divinylbenzene containing acid / basic functional groups)

❑ Softening of Water by Lime-Soda Method

The process of removing or reducing the concentrations of hardness causing ions/ salts from hard water is called as Softening of water.

Principle

In this method hard water is treated with calculated amounts of slaked Lime[Ca(OH)2] and soda ash Na2CO3 in reactions tanks. So as to convert hardness producing chemicals into insoluble compounds, which are then removed by settling and filtration.

Convert, Ca^{2+} in $CaCO_3$ and Mg^{2+} into $Mg(OH)_2$, Fe^{2+} → $Fe(OH)_2$

with Soda = Na_2CO_3 and Lime = $Ca(OH)_2$

Constituent	Reactions	Requirement L or S
1) Ca2+ (perm.) Eg. CaCl2, CaSO4	Ca2+ + **Na2CO3** → CaCO3 + 2Na+	**S**
2) Mg2+ (perm.) Eg. MgCl2, MgSO4	Mg2+ + **Ca(OH)2** → Mg(OH)2 + Ca2+ Ca2+ + **Na2CO3** → CaCO3 ↓ + 2Na+	**L + S**
3) HCO3- Eg. NaHCO3	2HCO3- + **Ca(OH)2** → CaCO3 ↓ + 2H2O + $CO3^{2-}$ = Na2CO3	**L – S**
4) Ca(HCO3)2 Temp.	Ca(HCO3)2 + **Ca(OH)2** → 2CaCO3 + 2H2O	**L**
5) Mg(HCO3)2 Temp.	Mg(HCO3)2 + **2Ca(OH)2** → Mg(OH)2 + 2 CaCO3 ↓ + 2H2O	**2L**
6) CO2	CO2 + **Ca(OH)2** → CaCO3 + H2O	**L**
7) H+	2H+ + **Ca(OH)2** → 2H2O + Ca2+ Ca2+ + **Na2CO3** → CaCO3 + 2Na+	**L + S**
8) FeSO4	FeSO4 + **Ca(OH)2** → Fe(OH)2 + CaSO4 Fe(OH)2 + O2 + H2O → Fe(OH)3 CaSO4 + **Na2CO3** → CaCO3 + Na2SO4	**L+S**

Constituent	Reactions	Requirement L or S
9) $Al_2(SO_4)_3$	$2Al^{3+} + 3 Ca(OH)_2 \rightarrow 2Al(OH)_3 + 3Ca^{2+}$ $3Ca^{2+} + 3Na_2CO_3 \rightarrow 3CaCO_3 + 6Na^{+}$	**L + S**
10) $NaAlO_2$	$NaAlO_2 + H_2O \rightarrow Al(OH)_3 + NaOH$	**-L**

Equivalence

$CaCO_3 = Ca(OH)_2 = Na_2CO_3$

100 = 74 = 106

Lime required = 74/100 [Mg^{2+} perm. + HCO_3^- + Ca^{2+} temp. + 2 Mg^{2+} temp. + CO_2 + H^+ + Fe^{2+} + Al^{3+} -$NaAlO_2$] mg/L

$$= \frac{74 [Mg^{2+} \text{ perm.} + HCO_3^- + Ca^{2+} \text{ temp.} + 2\ Mg^{2+} \text{ temp.} + CO_2 + H^+ + Fe^{2+} + Al^{3+} - NaAlO_2] \times \text{Volume of water} \times 100}{100 \times 1000 \times \% \text{ Purity}}$$

= g of Lime

Soda required = $\frac{106 [Ca^{2+} \text{ perm.} + Mg^{2+} \text{ perm} - HCO_3^- + H^+ + Fe^{2+} + Al^{3+}] \times \text{volume of water} \times 100}{100 \times 1000 \times \% \text{ purity}}$

= ___g of Soda

Non-hardness causing substances which doesn’t consumes lime – soda:

- KCl
- NaCl
- SiO2
- Na2So4
- Fe2O3
- K2SO4

∴ Lime requirement = ...

Soda requirement = 318 g

(ix) Calculate the quantity of lime and soda required to soften 20,000 liters of water containing following salts:

$CaCO_3$ = 10.0 mg/L $MgCO_3$ = 8.4 mg/L

$CaCl_2$ = 11.1 mg/L $MgSO_4$ = 6.0 mg/L

SiO_2 = 1.2 mg/L

assuming the purity of lime as 90% and soda as 95%.

Solution

Conversion to $CaCO_3$ equivalent

Constituent	Amount of substance (W)	Equivalent wt. of substance (E)	$CaCO_3$ equivalent = $\frac{W \times 50}{E}$
$CaCO_3$	10 mg/L	50	10 mg/L
$MgCO_3$	8.4 mg/L	42	10 mg/L
$CaCl_2$	11.1 mg/L	55.5	10 mg/L
$MgSO_4$	6.0 mg/L	60	5 mg/L
SiO_2	1.2 mg/L		Does not impart hardness

$$\text{Lime requirement} = \frac{74}{100}(\text{Temp. } Ca^{2+} + 2 \times \text{Temp. } Mg^{2+} + \text{Perm. } Mg^{2+}) \times \text{Volume of water} \times \text{purity factor}$$

$$= \frac{74}{100}[10 + 2 \times 10 + 5] \times 20{,}000 \times \frac{100}{90}$$

Lime requirement = 0.5755 kg

$$\text{Soda requirement} = \frac{106}{100}[\text{Perm.}(Ca^{2+} + Mg^{2+})] \times \text{Volume of water} \times \text{purity factor}$$

$$= \frac{106}{100}[10 + 5] \times 20{,}000 \times \frac{100}{95}$$

$$= 0.3347 \text{ kg}$$

... = 24 mg/L, CaS

Numerical 1: Calculate the amount of lime (74% pure) & Soda (90% Pure) for softening of 6000 litres of water containing following impurities.

$Mg(HCO_3)_2$ =73mg/L, $CaCl_2$ = 222mg/L, $MgSO_4$ = 120mg/L, $Ca(NO_3)_2$ = 164mg/L.

Soln

Impurities	Multiplication Factor	$CaCO_3$ eq.	Requirement
1) $Mg(HCO_3)_2$ = 73mg/L	100/ 146 or 50/73	= 73 x 50/73 =50	2L
2) $CaCl_2$ = 222mg/L	100/111 or 50/55.5	222 x 100/111 = 200	S
3) $MgSO_4$ = 120mg/L	100/120 or 50/60	120 x 100/120 = 100	L + S
4) $Ca(NO_3)_2$ = 164mg/L	100/164 or 50/82	164x 100/164 = 100	S

$$\text{Lime required} = \frac{74 \times [\text{CaCo3 eq of 2x } Mg(HCO_3)_2 + MgSO_4] \times \text{volume of water} \times 100}{100 \times 1000 \times \%\ \text{purity}}$$

$$= \frac{74 \times [\ 2 \times 50 + 100] \times 6000 \times 100}{100 \times 1000 \times 74} = 200 \times 6 = 1200g = 1.2Kg$$

$$\text{Soda Required} = \frac{106 \times [\ CaCO_3 \text{ eq of } CaCl_2 + MgSO_4 + Ca(NO_3)_2\ ] \times \text{vol. of water} \times 100}{100 \times 1000 \times \%\ \text{purity}}$$

$$= \frac{106 \times [\ 200+100+100] \times 6000 \times 100}{100 \times 1000 \times 90} = 2826.67g = 2.827Kg$$

Numerical 2: Calculate the quantity of soda and lime needed to soften 10000 litres of water containing following impurities. Na_2SO_4 =7mg/L, SiO_2 = 2mg/L, $CaCl_2$ = 22.2mg/L, $MgSO_4$ = 2.0mg/L, $Mg(HCO_3)_2$ = 5.0mg/L.

Soln:

Impurities	Multiplication Factor	$CaCO_3$ eq.	Requirement
1) Na_2SO_4 = 7mg/L	Non-hardness causing		
2) SiO_2= 2mg/L	Non-hardness causing		
3) $CaCl_2$ = 22.2mg/L	100/111 or 50/55.5	= 22.2 x 100/111 =20	**S**
4) $MgSO_4$ =2.0mg/L	100/120 or 50/60	= 2 x 100/120 =1.67	**L + S**
5) $Mg(HCO_3)_2$ = 5.0mg/L	100/146 or 50/73	= 5.0 x 100/146 = 3.42	**2L**

$$\text{Lime Required} = \frac{74 \times [CaCo_3 \text{ eq of } MgSO_4 + 2x\ Mg(HCO_3)_2 \times 10000}{100 \times 1000}$$

$$= \frac{74 \times [1.67 + 3.42x2] \times 10000}{100 \times 1000} = 62.974g$$

$$\text{Soda Required} = \frac{106 \times [CaCo_3 \text{ eq. of } CaCl_2 + MgSO_4] \times \text{vol of water}}{100 \times 1000}$$

$$= 229.702g$$

Numerical 2: Calculate the quantity of soda and lime needed to soften 10000 litres of water containing following impurities. Na_2SO_4 =7mg/L, SiO_2 = 2mg/L, $CaCl_2$ = 22.2mg/L, $MgSO_4$ = 2.0mg/L, $Mg(HCO_3)_2$ = 5.0mg/L.

Numerical 3: Calculate amount of lime (90% pure) and Soda (98% Pure)for the treatment of 1 million litres of water containing Ca(HCO3)2 = 8.1ppm, CaCl2=33.3ppm, HCO3-= 91.5ppm, MgCl2 = 38ppm, Mg(HCO3)2 = 14.6ppm. The coagulant Al2(SO4)3 was added at the rate of 17.1ppm.

Impurities	Multiplication Factor	CaCO3 eq.	Requirement
1) Ca(HCO3)2 = 8.1mg/L	= 100/162 or 50/81	5	L
2) CaCl2 = 33.3mg/L	100/111 or 50/55.5	30	S
3) HCO3- = 91.5mg/L	=100/122 or 50/61	75	**L - S**
4) MgCl2 =38.0mg/L	100/95 or 50/47.5	40	**L + S**
5) Mg(HCO3)2 = 14.6mg/L	100/146 or 50/73	10	**2L**
6) Al2(SO4)3 = 17.1mg/L	100/114 or 50/57	15	**L + S**

Lime l

Soda required = 10810.53g = 10.810kg

Types of Lime Soda Processes

- Hot Lime-Soda Method
- Cold Lime-Soda Method

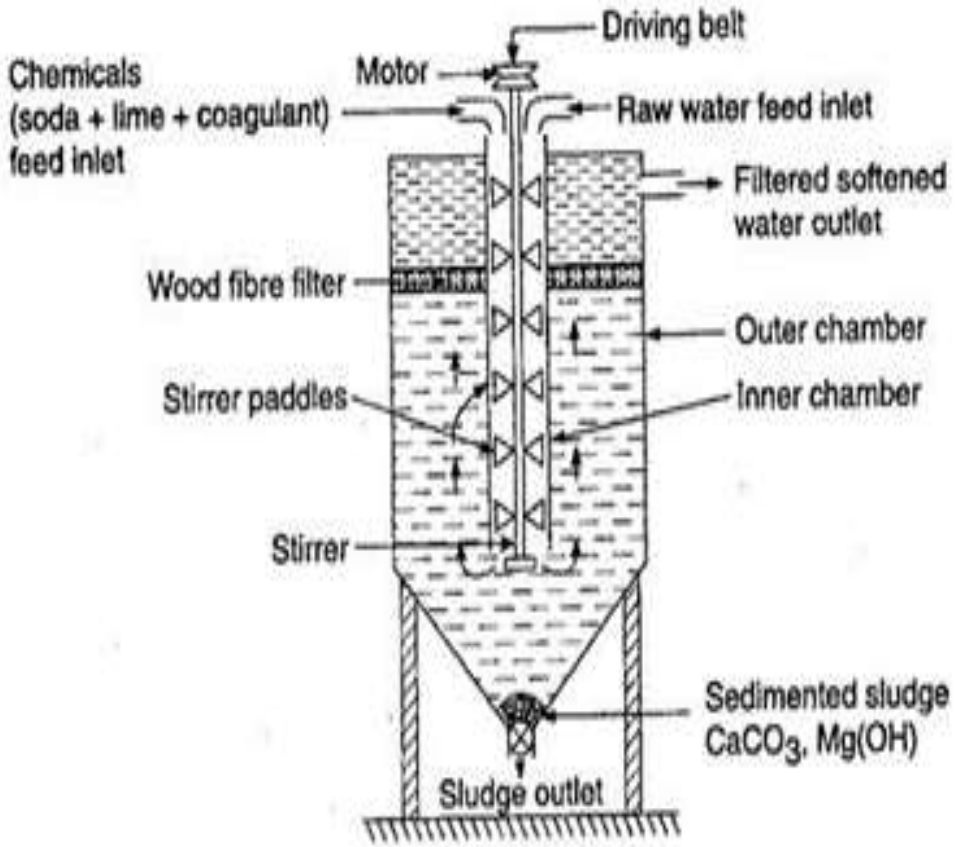
➢ Cold Lime Soda Method

Principle:

1. In this process calculated quantities of lime and soda are mixed with hard water at room temperature.
2. The **precipitate** formed are finely divided and there fore do not settle down easily and cant be filtered easily.
3. Thus it is essential to add small quantities of coagulants like alum, aluminum sulphate, sodium aluminates.

These hydrolyze to **gelatinous ppt of aluminum hydroxide** and entrap fine precipitate.

If sodium aluminate is used as coagulant, silica as well as oil present in water are also removed.



Cold lime soda process provides water containing a Residual hardness of 50 to 60 ppm.

$NaAlO2 + 2H2O \rightarrow NaOH + Al(OH)3$

$Al2(SO4)3 + 3Ca(HCO3)2 \rightarrow 2Al(OH)3 + 3CaSO4 + 6CO2$

➢ Hot Lime Soda Method

Principle:

1. In this method water is treated with softening chemicals like soda and Lime at a **temp. of 80°C to 150°C**. this process is operated at a temp close to boiling point of the solution.

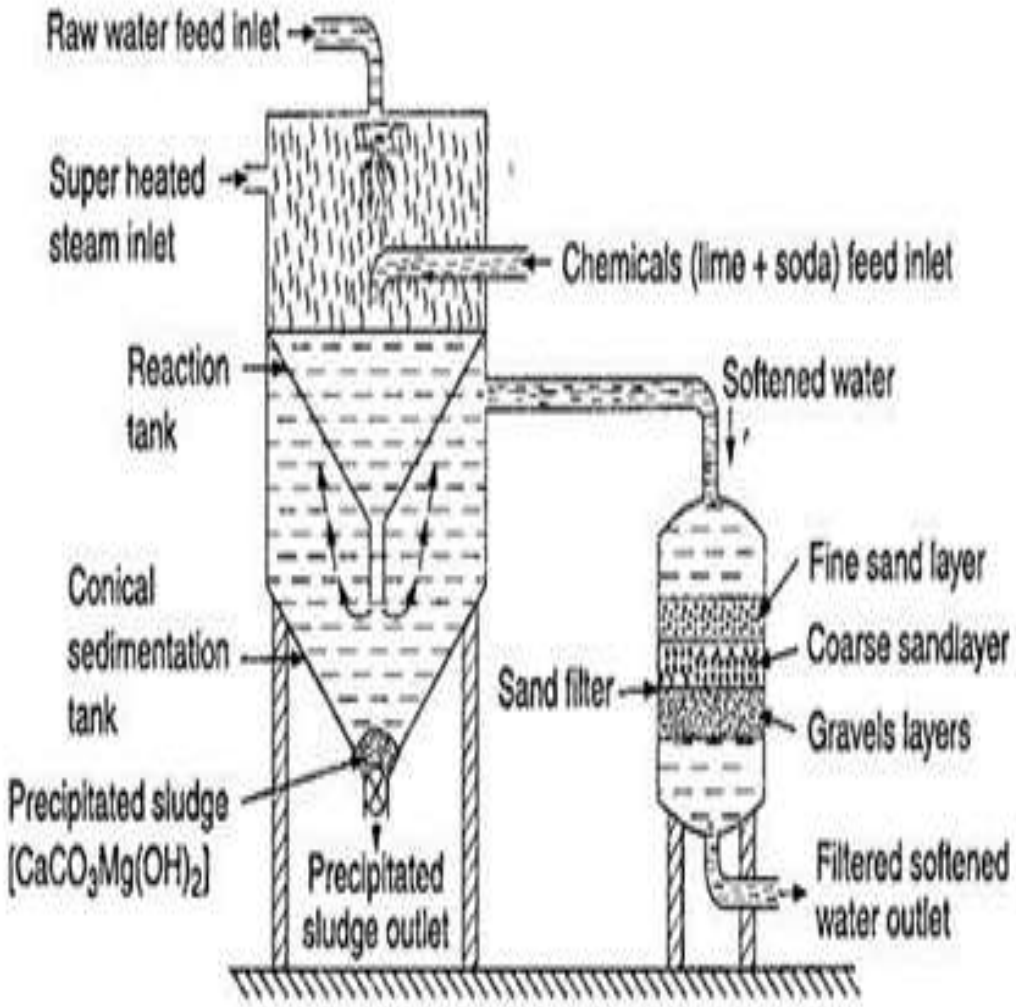
Hence the reaction proceeds faster, the softening capacity.

2. Of the process increased many fold, the precipitate and sludge formed settle down rapidly and hence no coagulants are required.

Much of the dissolved gases like CO2 & air are driven out of water.

Viscosity of soften water is lowered therefore filtration of water becomes much easier.

This method produces water of comparatively lower residual hardness of 15 to 30 ppm.



Advantages of Lime Soda Process

- It is very Economical
- If the process is combined with sedimentation & coagulation, lesser amounts of coagulants shall be needed.
- The process increases the pH value of the treated water, thereby corrosion of the distribution pipe is reduced.
- Besides the removal of hardness, the quantity of minerals in the water is reduced.
- To a certain extent, iron and manganese are also removed from the water.
- Due to alkaline nature of treated water, the amount of pathogenic bacterial in water is considerably reduced.

Disadvantages

- For efficient and economical softening, careful operation & skilled supervision is required.
- Disposal of large amounts of sludge poses a problem. However the sludge may be disposed off in raising low lying areas of the city.
- This can remove hardness only up to 15ppm which is not good for boilers.

Softening of water by zeolite process

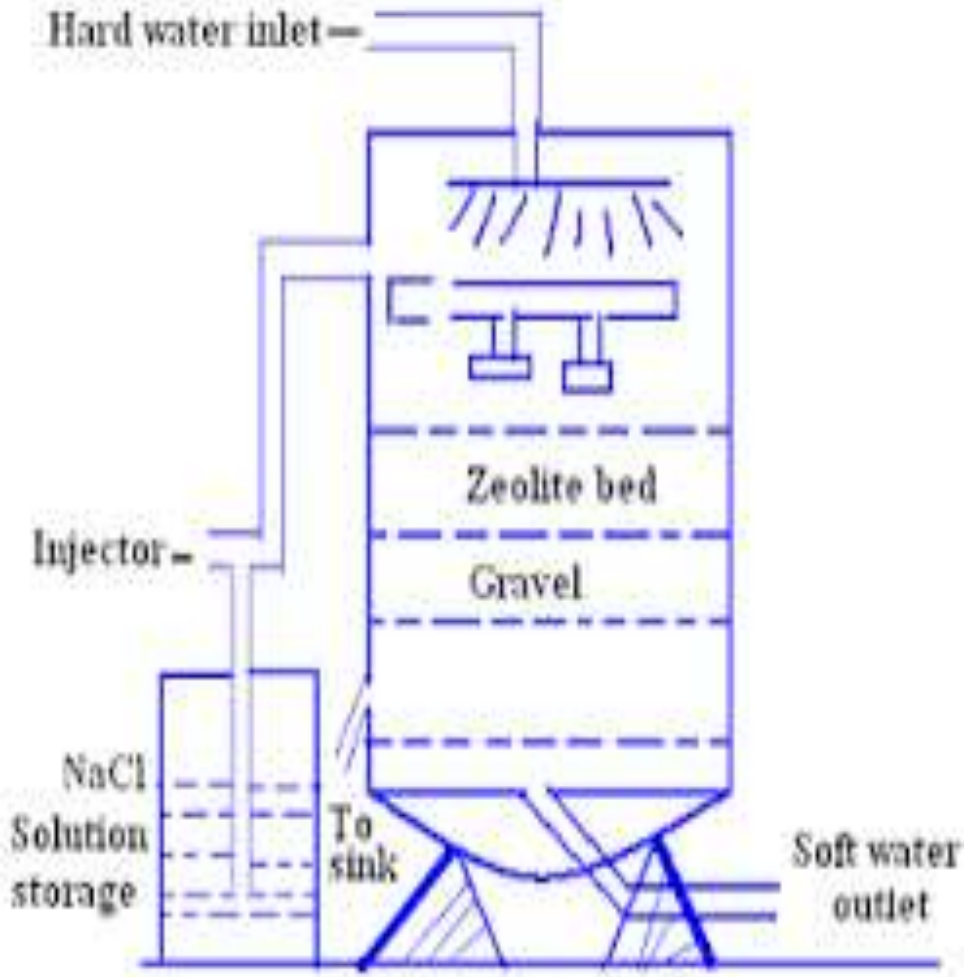
- Sodium zeolite: $Na_2O.\ Al_2O_3.xSiO_2.yH_2O$
- Where x = 2-10 and y = 2-6
- Zeolite is sodium alumino silicate capable of exchanging reversibly its sodium ions for hardness producing ions present in water.

❑Types of zeolite

- Natural Zeolite: are non porous eg. Natrolite $Na_2O.Al_2O_3.4SiO_2.2H_2O$
- Synthetic Zeolite: are porous and possess gel structure. They are prepared by heating together china clay, feldspar and soda ash.



Natrolite is a fibrous, sodium-rich zeolite mineral, a hydrated sodium and aluminum silicate with the chemical formula $Na_2Al_2Si_3O_{10}\cdot 2H_2O$



Softening Process :

$$Na_2ze + Ca(HCO_3)_2 \longrightarrow CaZe + 2\ NaHCO_3$$
$$Na_2ze + Mg(HCO_3)_2 \longrightarrow MgZe + 2\ NaHCO_3$$

Regeneration :

$$CaZe + 2\ NaCl \longrightarrow Na_2ze + CaCl_2$$
$$MgZe + 2\ NaCl \longrightarrow Na_2ze + MgCl_2$$

Advantages of zeolite process

- It removes the hardness almost completely and water of about 10ppm hardness is produced.
- The equipment used is compact, occupying small space.
- No impurities are precipitated, so there is no danger of sludge formation in treated water at a later stage.
- The process automatically adjusts itself for variation in hardness of incoming water.
- It is quite clean process.
- It requires less time for softening.
- It requires less skill for maintenance as well as operation.

Disadvantages

- The treated water contains more sodium salts than in lime soda process.
- The method only replaces Ca2+ and Mg2+ ions by Na+ ions but leaves all the acidic ions like HCO_3- and $CO_3$2- as such in the softened water.
- When such soften water is used in boilers for steam generation, sodium carbonate decomposes producing CO2, which cause corrosion and sodium carbonate hydrolyses to sodium hydroxide which cause caustic embrittlement.
- High turbidity water cannot be treated efficiently by this method.

Numerical 1: The hardness of 10000 litres of hard water sample was completely removed by passing it through a zeolite softener. The zeolite softener required 5000litres of NaCl soln containing 1170mg of NaCl/L. Determine hardness of water sample.

Solution:

5000 L of NaCl soln = 5000 x 1170mg of NaCl

= 5850000mg of NaCl

1eq of NaCl = 1eq of $CaCO_3$

58.5mg of NaCl = 50mg of $CaCO_3$

5850000mg of NaCl = $\frac{5000000}{}$ mg of $CaCO_3$

There fore,

10000L of Hard water = 5000000mg of $CaCO_3$

1L of Hard water = 5000000/10000 = 500mg of $CaCO_3$

Hence Hardness of water = 500ppm.

Numerical 2: An Exhausted zeolite softener was regenerated by passing 75L of NaCl soln having strength of 75g/L. If 1.6×10^4 hard water was passed though zeolite bed, what will be the hardness of water.

Soln:

75L of NaCl = 75 x 75 g of NaCl = 5625g of NaCl

= 5625×10^3 mg of NaCl

1eq of NaCl = 1eq. Of CaCO3

58.5mg of NaCl = 50mg of CaCO3

5625×10^3 mg of NaCl = $5625 \times 10^3 \times 50/58.5$ mg of $CaCO_3$

= 4.81×10^6 mg of $CaCO_3$

1.6×10^4 L of hard water = 4.81×10^6 mg of CaCO3

1L of hard water = $4.81 \times 10^6 / 1.6 \times 10^4$

= 300ppm

Numerical 3: An exhausted zeolite softener was regenerated by passing 125L of NaCl having strength 50gNaCl/L. If the hardness of water is 350ppm, then calculate the total volume of water sample.

Soln:

125L of NaCl = 125 x 50g of NaCl

= 6250g = 6250000mg of NaCl

58.5mg of NaCl = 50mg of $CaCO_3$

6250000mg of NaCl = 6250000x50/58.5

= 5341880mg of $CaCO_3$

1L of hard water = 350mg of $CaCO_3$

Therefore,

x L of Hard water = 5341880mg of $CaCO_3$

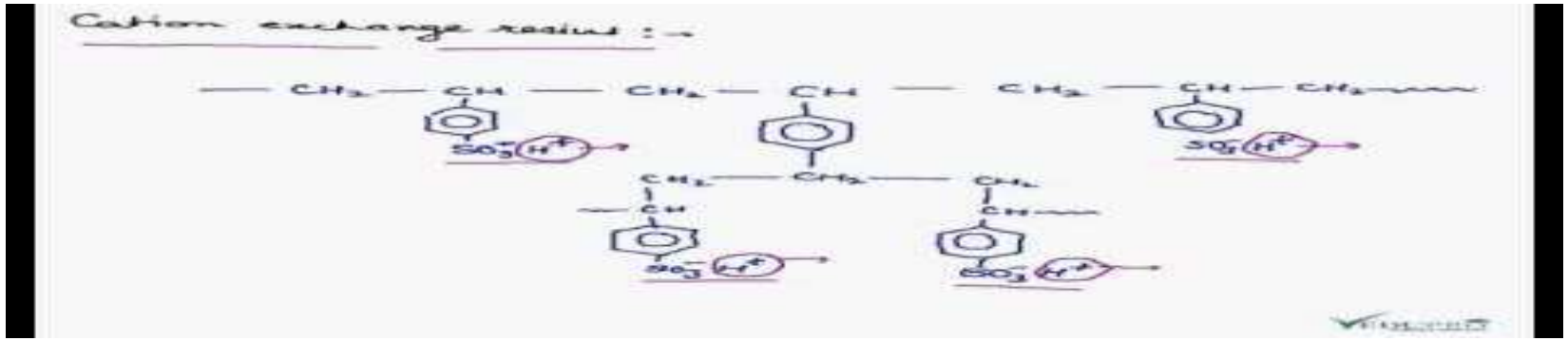
Hence volume of hard water = 5341880/350

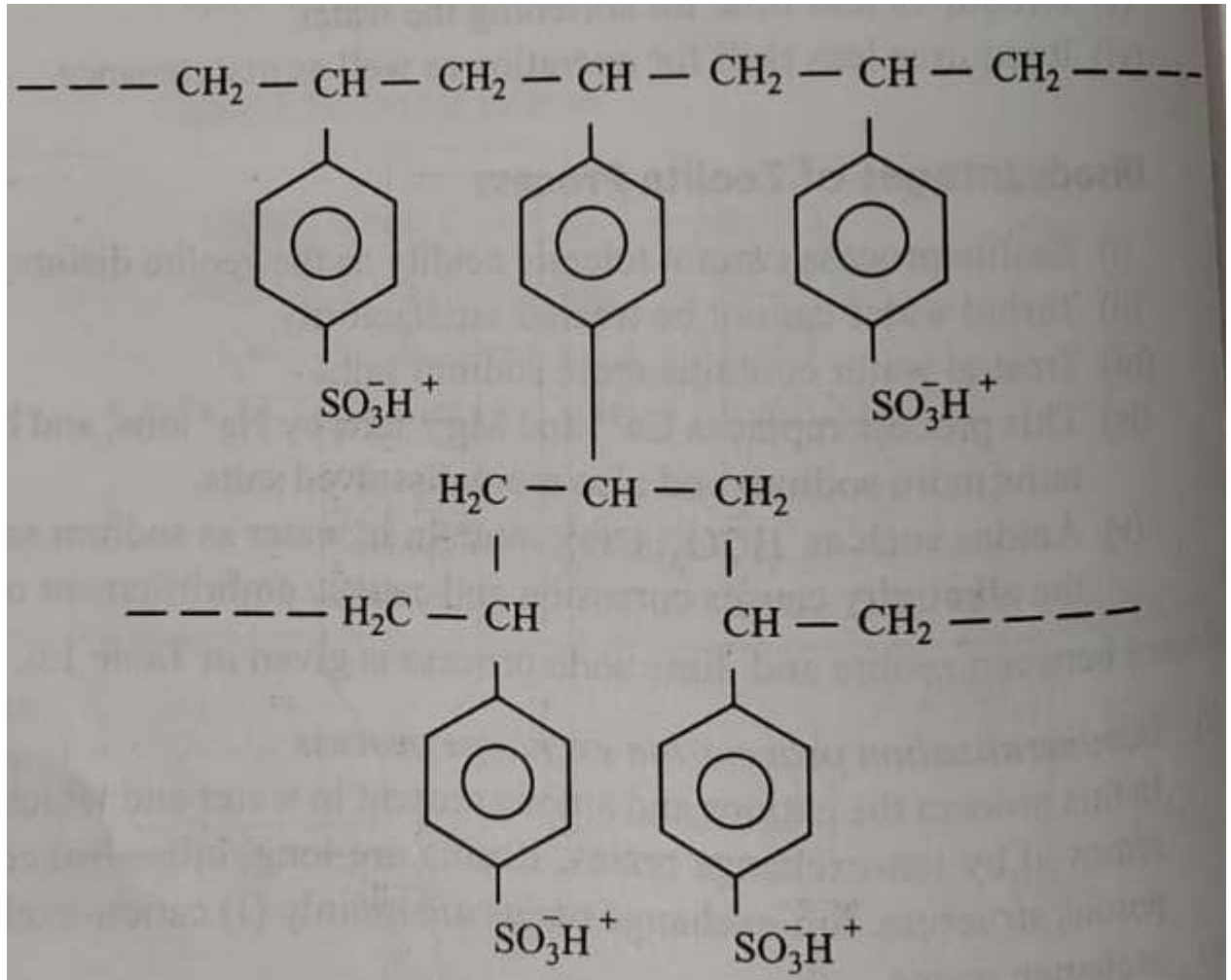
= 15262.5L

- Numerical4: Calculate the volume of NaCl (12.5g/L) in L for regeneration of zeolite, the 200ppm of 10000L of hardwater was passed through it to get zeolite exhausted.

Softening of Water by Ion-Exchange Process Or Demineralization process

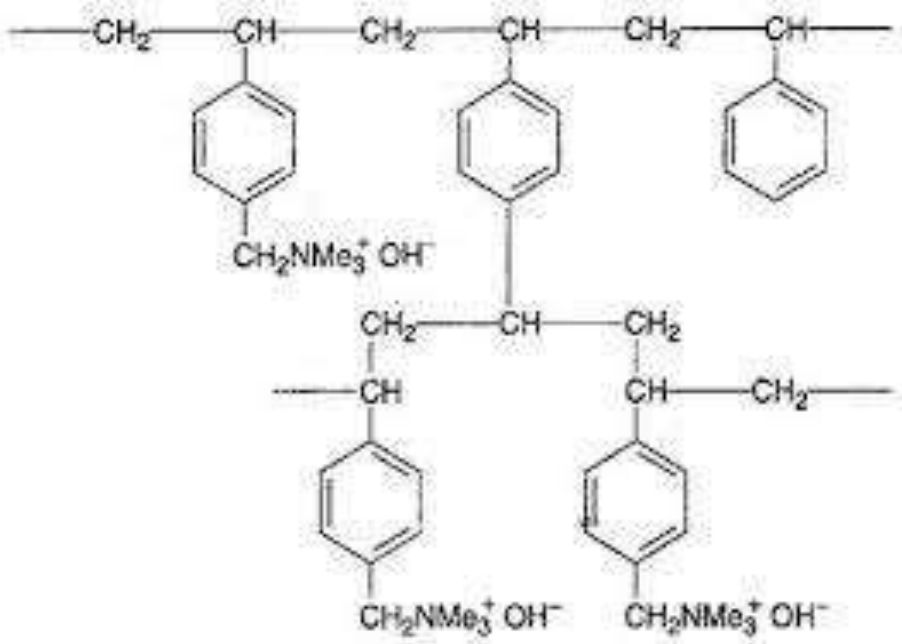
- Ion exchange resins are insoluble cross linked long-chained organic polymers with a microporous structure.
- **Cation Exchange resin:** Styrene-divinyl benzene copolymers which on sulphonation or carboxylation become capable of exchanging their hydrogen ions with the cations in hard water.

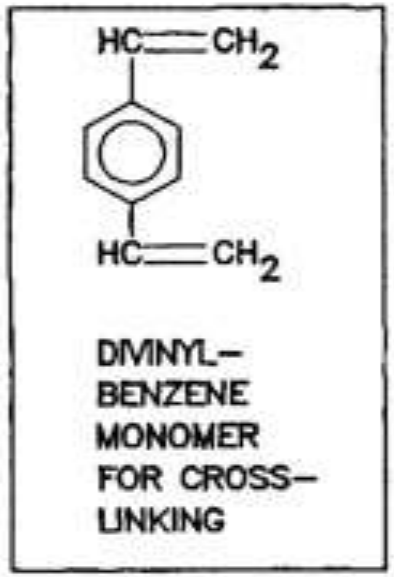


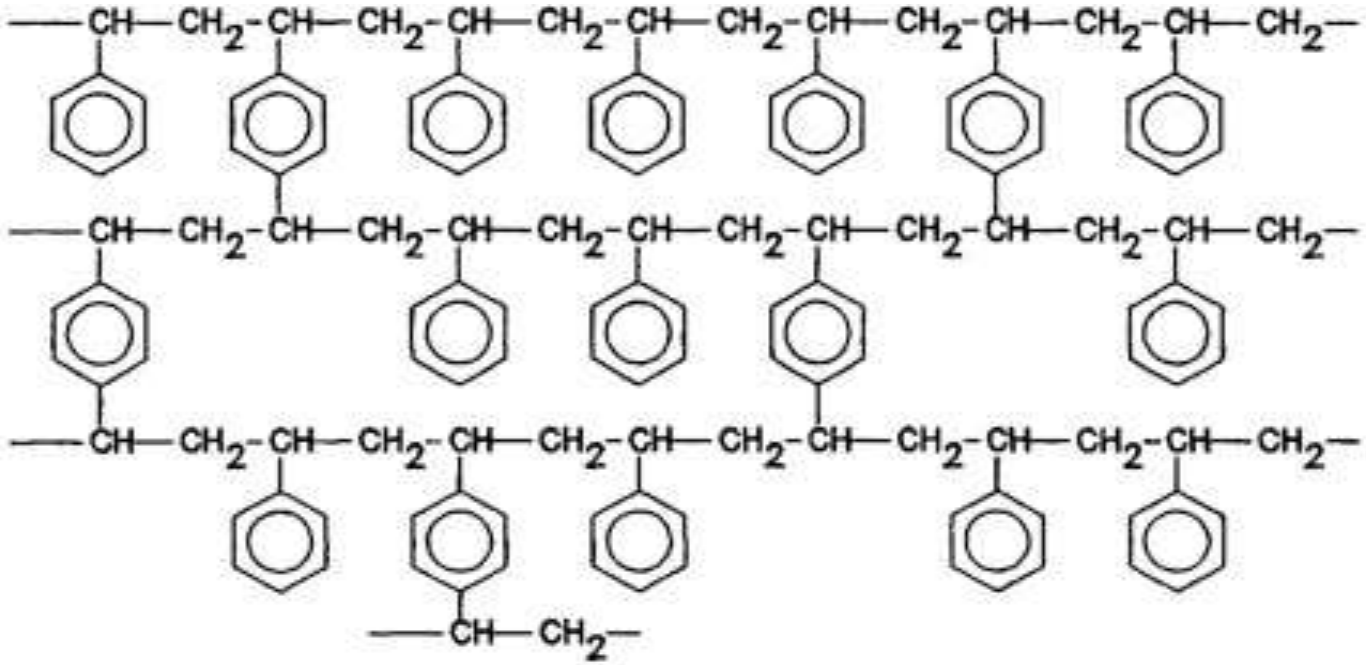


Sulphonation form of cation-exchange resin

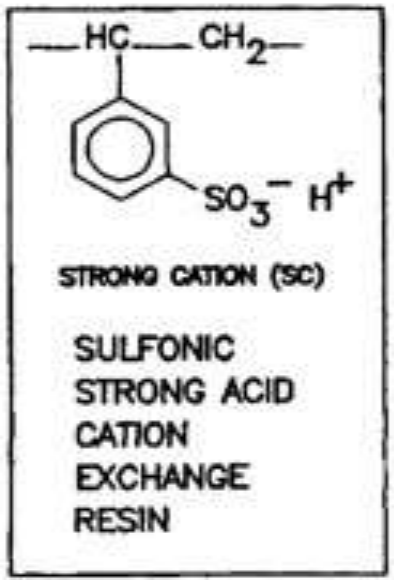
Anion Exchange resin: Styrene-divinyl benzene copolymers which on further functionalization like formylation followed by Mannich reaction or nitration followed by amination and alkylation gives quaternary ammonium groups as an integral part of the resin matrix. These after treatment with dil NaOH soln become capable of exchanging their OH- anions with anions of hard water.

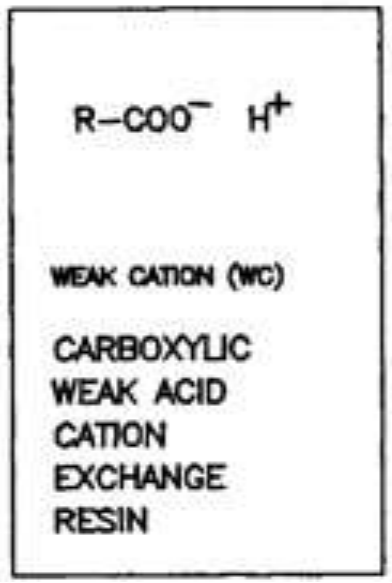


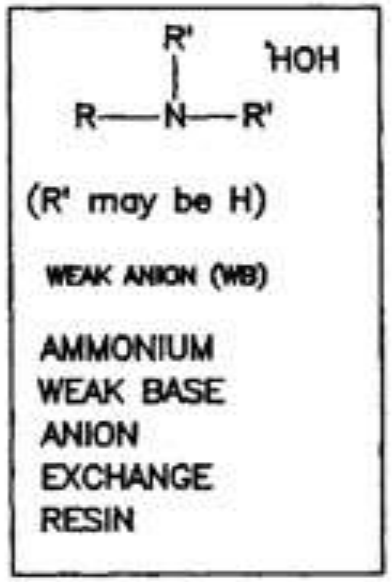


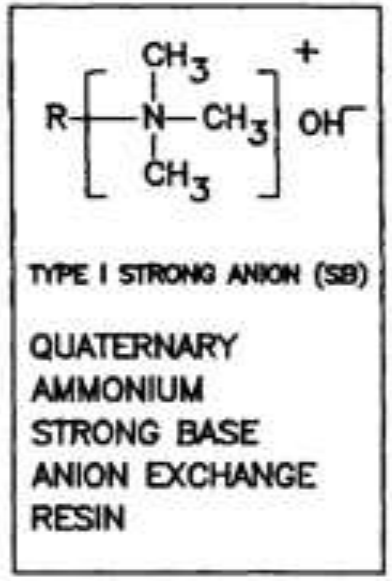


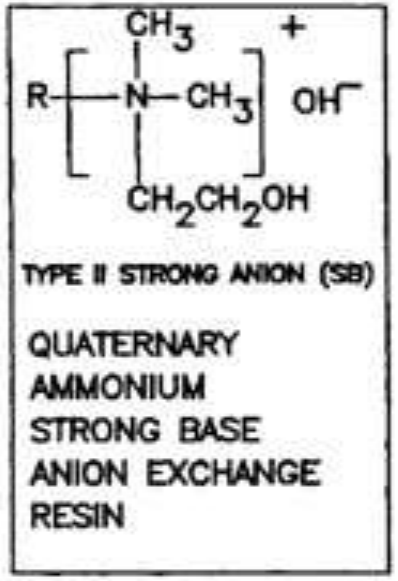
POLYSTYRENE DIVINYLBENZENE BACKBONE



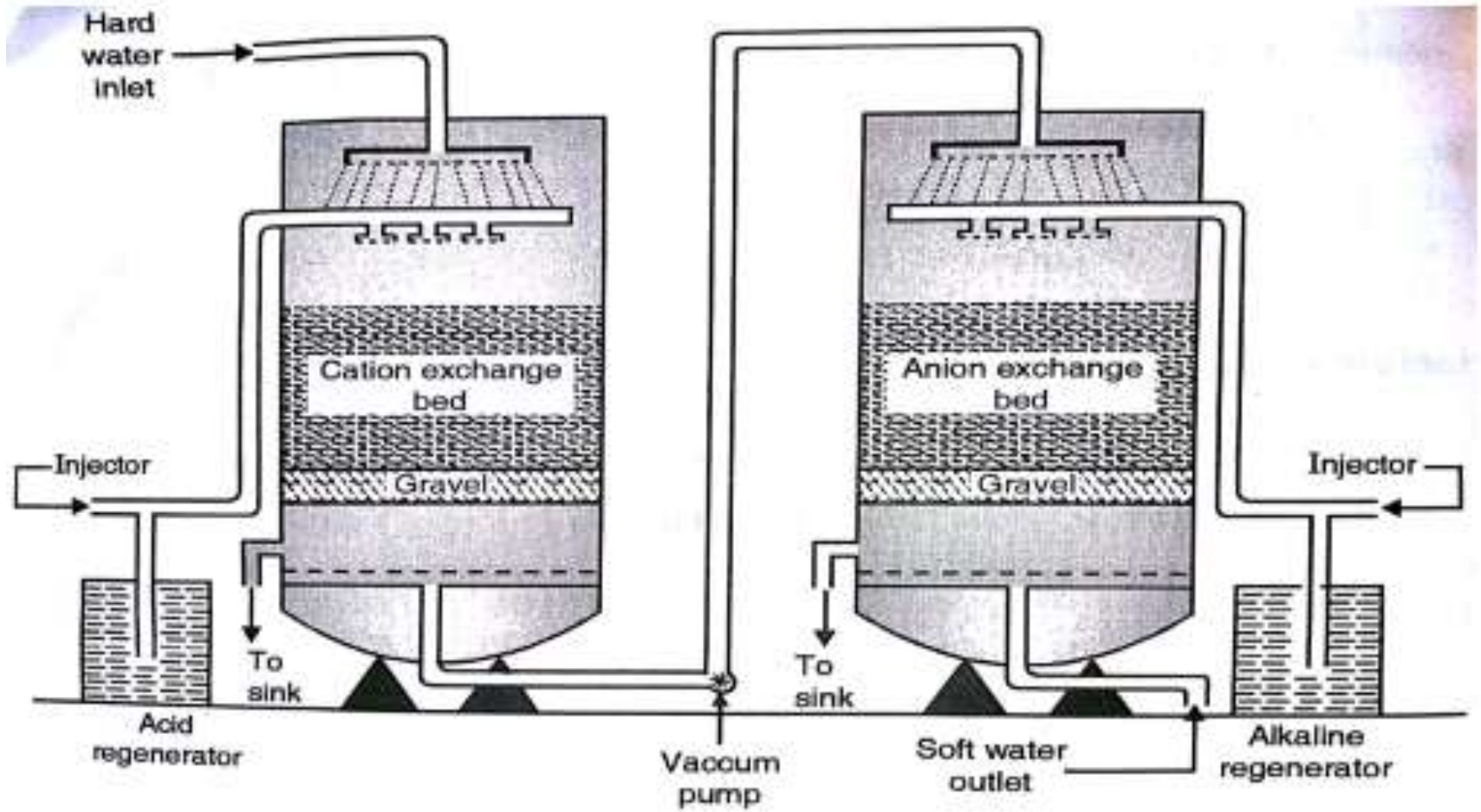








Process: $Ca(HCO_3)_2$, $CaCl_2$, $MgCl_2$



Demineralisation Diag. 2

Reactions with Cation Exchanger: RH+

$RH^+ + Ca^{2+} \rightarrow R_2Ca + 2H^+$

Resin Hardwater Resin Soften water

$RH^+ + Mg^{2+} \rightarrow R2Mg + 2H^+$

HCO_3^-, Cl^-, SO_4^-

Reactions with Anion Exchanger: R'OH

$R'OH + Cl^- \rightarrow R'Cl + OH^-$

Resin Hard Water Resin Soften water

$2R'OH + SO_4^- \rightarrow R'2SO4 + 2OH^-$

Overall Reaction:

$H^+ + OH^- \rightarrow H2O$

Regeneration Reactions

Cation Exchanger

$R_2Ca + 2HCl \rightarrow RH^+ + CaCl2$

Regenerated Resin Washings

Anion Exchanger

$R'_2SO_4 + 2NaOH \rightarrow 2\ R'OH + Na_2SO_4$

➢ Advantages

- The process can be used to soften highly acidic or alkaline water.
- It produces water of very low residual hardness (2ppm) so it is very good for treating water for use in high pressure boilers.

➢ **Disadvantages:**

- The Equipment is costly and more expensive chemicals are needed.
- If water contains turbidity then the output of the process is reduced. The turbidity must be below 10ppm. If it is more it has to be removed first by coagulation and filtration.

Numerical 1: After treating 10000 litres of water by ion exchanger the cationic resin required 200 litres of 0.1N HCl and anionic resin required 200 Litres of 0.1N NaOH soln. Find the hardness of water.

Soln:

Hardness of 10000L of water = 200 L of 0.1N HCl

= 200L of 0.1 N NaOH

1 eq. of HCl = 1 Eq. of $CaCO_3$

0.1N of HCl = 0.1N of $CaCO_3$

Hence,

Hardness of 10000L of water = 200L of 0.1N $CaCO_3$ eq.

= 20L of 1N $CaCO_3$ eq.

Std Statement: 1L of 1N $CaCO_3$ = Eq. wt = 50g

Hardness of 10000L of water = 20 x 1L of 1N $CaCO_3$

= 20 x 50 g = 1000g

= 1000000mg of $CaCO_3$ eq.

Hence,

Hardness of 1L of water = 1000000/10000 = 100mg/L = 100ppm

- Numerical 2: After treating 50000 litres of water by ion exchanger the cationic resin required 150 litres of 0.25N HCl and anionic resin required 150 Litres of 0.25N NaOH soln. Find the hardness of water.

BOD:

Biological oxygen demand of sewage is defined as the amount of free oxygen required for the biological oxidation of the organic matter under aerobic condition at 20°C for a period of 5days.

Unit of BOD = ppm or mg/L

- Determination of BOD

1. A known volume of effluent sample is diluted with a known volume of water. Sample is diluted to ensure that sufficient oxygen is available for the complete aerobic oxidation of organic matter.
2. The diluted sample is divided into two parts and kept in two stoppered bottles. The dissolved oxygen content of one of the bottle is determined by Winkler's method, which is blank reading. It is denoted as DO_b .
3. Another bottle is incubated at 20°C for 5days. Then dissolved oxygen is determined. This is denoted as DOi.

The difference between DOb and DOi gives BOD.

BOD = (DOb-DOi) x dilution factor

= (DOb-DOi) x Volume of sample after dilution/volume of sample before dilution.

BOD test is usually influenced by the type of micro-organisms, presence of toxins, pH, Some reduced minerals; etc.

Reaction:

Organic Matter + O_2 → CO_2 + H_2O

BOD is proportional to the amount of organic waste to be degraded.

An average sewage has BOD of **100-150** ppm.

Drinking water should have BOD less than 1ppm.

Significance of BOD:

1. It indicates the amount of decomposable organic matter in sewage.
2. The higher is the BOD of a sample the higher the amount of decomposable organic matter in the sample and the higher is the pollution in water.
3. From BOD self purifying capacity of streams cans be determined. Thus it provides the degree of pollution at any time in the sewage stream.

COD

Chemical oxygen demand may be defined as the amount of oxygen required for the oxidation of chemically degradable organic matter in the hot conditions with the use of oxidants like acidified $K_2Cr_2O_7$ in 2-3hrs.

Unit of COD = mg/L or ppm

- Determination of COD:

1. A known volume of waste water sample is refluxed with a known excess of standard potassium dichromate soln and dilute sulphuric acid in presence of silver sulphate for 2-3hs. Silver sulfate is used as catalyst to promote oxidation of straight chain aliphatic compounds, aromatic compounds.
2. The organic matter of the sample is oxidized to ammonia, CO_2 and H_2O. The unreacted $K_2Cr_2O_7$ is titrated against Ferrous ammonium sulphate[$FeSo_4.(NH_4)_2SO_4.H_2O$]i.e. Mohr's Salt.
3. This experimentally measured amount of potassium dichromate is used to calculate the equivalent oxygen required by the waste water for degradation of the pollutants.

Reactions

$C_xH_yO_z + (x + y/4 - z/2)\ O_2 \rightarrow x CO_2 + y/2\ H_2O$

The unreacted dichromate solution is titrated against Mohrs salt using ferroin indicator.

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

$[Fe^{2+} \rightarrow Fe^{3+} + e] \times 6$

Overall reaction

$Cr_2O_7^{2-} + 14H^+ + 6\ Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$

The COD is calculated as

$$COD = \frac{(V1-V2) \times N_{FAS} \times 8000}{\text{Volume of sample}}$$

V1 = vol. of FAS in blank titration

V2 = vol. of FAS in back titration

Significance of COD

The COD value is not affected by the presence of toxins and other unfavaourable conditions for the growth of microorganisms.

It is taken as basis for calculation of efficiency of treatment plant.

It help in designing the water treatment plant.

Numerical 1: A 50mL of sample contains 840ppm of dissolved oxygen. After 5days the dissolved oxygen value becomes 230 ppm, after the dilution of sample to 80mL. Calculate the BOD.

Soln: BOD = (Dob-Doi) x Volume of sample after dilution/ vol of sample before dilution

BOD = (840-230) x 80/50 = 976ppm

Numerical 2: A 25mL of sewage water sample was refluxed with 10mL of 0.25N $K_2Cr_2O_7$ soln in presence of dil.H_2SO_4, Ag_2SO_4 and $HgSO_4$. The unreacted dichromate required 6.5ml of 0.1N FAS. 10mL of same $K_2Cr_2O_7$ required 27mL of 0.1N FAS. Calculate COD.

Given: Volume of sample = 25mL

Blank titration reading (V1) = 27mL

Back titration reading (V2) = 6.5mL

COD= (V1-V2) x N(FAS) x 8000/Vol. of Sample

= (27-6.5) x 0.1 x 8000/25 = 656ppm

Numerical 3: A 30mL of sewage water sample was refluxed with 15mL of 0.25N $K_2Cr_2O_7$ soln in presence of dil.H_2SO_4, Ag_2SO_4 and $HgSO_4$. The unreacted dichromate required 14 ml of 0.05N FAS. 15mL of same $K_2Cr_2O_7$ required 20mL of 0.1N FAS. Calculate COD.

Note: Covert one statement in to fixed normality.

Volume of sample = 30mL

Blank Titration reading (V1) = 20ml of 0.1N FAS = **40 mL** of **0.05N** FAS

Back Titration reading (V2) = **14mL** of **0.05N** FAS

COD = 346.667ppm